



K-Means Clustering Transcript

Hello everyone, welcome to lesson 1 video 3. In this video, we'll learn about k-means clustering, a widely used algorithm for partitioning data into clusters. Now, within clustering, as I've told you, there are various different types of algorithms that we use. One type is what you saw as agglomerative clustering, wherein you use dendograms to find out similar data points and then we, based on that similarity, club them together into a similar group.

We start with n different groups and then we go towards forming one single group, okay. Now, in k-means, this is again one other algorithm which we use for grouping data points together, which is called as k-means. In k-means, our idea is to partition entire data or the entire space into different groups, okay.

And how does this entirely work is what has been guided here, okay. So this is a feature space partitioning algorithm, okay. It doesn't work like how hierarchical clustering works, but it works on finding similarities between data points and then clubbing them together based on finding a group or the centroid, okay.

Once you have found the centroid of the group, you can club all the data points which are nearest to that centroid into one group and then you keep on changing the centroid unless and until the algorithm doesn't change or the centroid doesn't change its position. What do I mean by all of this is what I'll explore on the website I have ahead or we had created for you. So let's say here, you see that on this page you have all the data points, right.

This is where you will learn about how to interactively play with k-means clustering. So initially, the algorithm starts with asking you how many clusters do you want, okay. You can say 1 group, 2 group, 3 group, 4 group, 5 group or whatever number of groups, okay.

This is just for demonstration, hence I have not included more than 5. So let's say I would say 1 group in that case. So if I say 1 group, every data point is in 1 group which is not what we need, right. So let's say if I start with 2 groups and I click initialise, now you see that you start randomly by choosing 2 centroids.

So herein you have 1 centroid which is looking like a star here and then you have the other centroid. Now what you do in the next step is to find out which particular data points are closure and club them together in that group, okay. Now you see that all of these data points are closure, hence they are part of orange group.

These data points are closure, hence they are part of blue group. So whatever is closure to blue star are blue group, whatever is closure to orange star are orange group. Now you see that your entire space is partitioned and that's what is meant by partitioning the space.

So you are creating divide in the space, the regions in the space as group 1 or the region which belongs to group 2 and so on. Now in this case if a new observation comes in, let's say herein there is a new observation, a new individual or new data point arrives. Because this data point resides in the region of orange, you would say that this data point is actually belonging to orange cluster and that's the entire essence of k-means clustering.

Now this, once you have assigned the clusters, it does not mean that this is where you have optimally find the centroids, okay. So you update the centroids, you take mean of all of them, you see that centroid has gone to a new position. Now you keep updating the centroid based on let's say whatever data points are near.

If you see here, these data points were classified in the initial first step as orange. Now because they are closest to blue, they would be tagged as blue, okay. So in the next step, you would see that this is what happens, okay.

So this is how centroid keeps on changing its position and once it finds out that the centroid positions are not changing, that's when the algorithm converges. So if you see on the right hand side, it says that the algorithm has converged. Similarly let's say we start with number of clusters as 3. So we initialise or randomise three different points as clusters or random clusters.

Now we start taking distances of each data point from blue and similarly all the data points. Now if you think of this point, okay, if you concentrate on this point, the distance of that point from blue star and from orange star and from red star, it is very closest to blue one, right. So it would be tagged into a blue cluster and that's what you see that all of this, everything which is left to on the left side of this line, right, or this partition as all blue space or the blue category or blue group region, right.

So you assign them together into those groups and then you update the centroid, update as it, you take the mean of all the data points and you have a new position altogether, okay. So when I say mean of all the data points, because each of these, if you think of this is a two dimensional data, right. So each data point is a combination of X and Y coordinates.

So you'll take mean of X, mean of Y and hence you will get mean of X and Y as a new coordinate for each one of them. Then you update centroid, keep on updating it. So for full demonstration, let's see here, right, you see how the centroids keeps changing its position.

Now you see that the centroid keeps changing its position and once the algorithm has converged, you say that the algorithm, because the centroids are not changing its position, hence the algorithm has converged. At the bottom, you, there is something which is written as within cluster sum of squares. This is what measures homogeneity within the cluster.



So let's say if within cluster sum of squares is too high, I mean, if it is too, too much, it means that there are no groups which which looks identical. This has to be small. Now, in this case, if you think of the data points are all scattered, right, it is not like they all are concentrated at one place and you have another group which is all concentrated, right? So if there are no specific distinction between the clusters, between the data points, you see that the entire data is randomly distributed here, right? Hence, within cluster sum of squares would be high.

This metric needs to be low for the clusters to be uniform or for the clusters to be homogeneous within them, right? So this is what I just wanted to introduce you with. Now, this is the same demonstration here, so I'll skip that. One of the good things about k-means clustering is it is fast and scalable.

It is easy to understand as well, right? But the weaknesses are you have to initially start with the number of clusters. Now, in your data, you might have two optimal clusters or ten optimal clusters. You don't know.

So how do you find out how many clusters actually exist in the data? That's what we'll learn in the next video. But at least the weakness of the algorithm right now is you do not know how many clusters to start with. Okay, because let's say if you have high dimensional data, two dimensional data is what you were able to visualise in this example.

Let's say you have one group here, one group here, one group here. But you are able to visualise it because your data is represented in two dimensions. But when your data is in ten dimension, you will not be able to visualise it.

In that case, how would you say whether there exist four groups or five groups or ten groups or whatever, right? So that's the main weakness of the algorithm. Then it is also sensitive to initial placement. So let's say you have chosen randomly the initial starting point of the centroids.

You choose three as the clusters, number of clusters, and then you have three starting points as the centroids. So if your choice becomes very random, I mean, then there are a few problems. One is it might take a long to converge.

It might not even converge as well to the local optimum. So you might have a very bad looking clusters as well. OK, it also assumes the clusters to be spherical in shape.

Let's say, for example, you have dense looking clusters which are spherical and they are of similar size. So that's also one of the things that it looks like because it is dependent on distances. OK, so let's learn about how to find k or the number of groups that you are looking for in clustering in the next video.

In summary, k-means clustering is a fast, efficient method for partitioning data into groups. Although it has some limitations, it remains one of the most widely used



clustering algorithm. In the next video, we'll discuss how to choose the right number of clusters.

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